1. Definitions and timeline

ILAE 2015

- t1 = 5 min (treat), t2 = 30 min (risk of damage) for convulsive SE
- Established SE: fails benzodiazepines
- **Refractory**: fails benzodiazepine + one second-line drug
- **Super-refractory**: continues ≥ 24 h after anaesthetic or recurs on weaning

Key point: Time is brain: every step has a time target.

2. First line: benzodiazepines

Doses

- Lorazepam 0.1 mg/kg IV (4 mg), repeat once
- Midazolam 10 mg IM/buccal (RAMPART: IM midazolam ≥ IV lorazepam pre-hospital)
- Diazepam 10 mg IV

Pitfalls

- Under-dosing is common; ABC, glucose (thiamine first if malnourished), check airway

Key point: Adequate benzodiazepine dose, then move on within 10 minutes.

3. Second-line agents

ESETT 2019

- Levetiracetam 60 mg/kg (max 4.5 g), fosphenytoin 20 mg PE/kg, valproate 40 mg/kg (max 3 g): **equal efficacy ≈ 47%**

Choice

- Levetiracetam: few interactions; adjust in renal failure
- Phenytoin: arrhythmia, hypotension, purple glove; monitor ECG; level
- Valproate: avoid in liver disease, mitochondrial disease, pregnancy; hyperammonaemia; levels
- Lacosamide 200–400 mg: PR prolongation; phenobarbital 15–20 mg/kg

Key point: No winner in ESETT: choose by comorbidity and side-effects.

4. Refractory SE: anaesthetic infusions

Agents

- **Midazolam** 0.2 mg/kg then 0.05–2 mg/kg/h: tachyphylaxis
- **Propofol** 1–2 mg/kg then 2–10 mg/kg/h: PRIS (keep < 5 mg/kg/h, < 48 h; CK, TG, lactate)
- **Ketamine** (NMDA): useful later when GABA receptors internalised; haemodynamically stable
- Thiopental/pentobarbital: hypotension, immunosuppression, long half-life

Goals

- **Continuous EEG**: seizure suppression or burst suppression for 24–48 h, then slow wean
- Intubation, vasopressors, temperature control

Key point: Balance seizure control against deep sedation harms: aim for the least EEG suppression that works.

5. Super-refractory SE and NORSE

Options

- Ketamine, ketogenic diet, immunotherapy (steroids, IVIG, plasma exchange, rituximab), magnesium, pyridoxine in children
- Hypothermia (HYBERNATUS: no outcome benefit), surgery for focus, ECT

NORSE / FIRES

- New-onset refractory SE without clear cause; autoimmune encephalitis (NMDA-R) common
- Early first-line immunotherapy within 72 h; anakinra/tocilizumab in FIRES

Key point: Think autoimmune early in previously healthy young adults.

6. Diagnostic work-up for aetiology

Tests

- Glucose, Na, Ca, Mg, urea, liver tests, ammonia, toxicology, antiseizure drug levels, pregnancy (eclampsia)
- CT then MRI; LP (infection, autoimmune panel, HSV PCR)
- EEG to confirm cessation and detect non-convulsive SE

Common causes

- Low drug levels/withdrawal, alcohol, stroke, tumour, trauma, infection (HSV), autoimmune, metabolic, eclampsia

Key point: Treat seizures and search for the cause at the same time.

7. Other neuro emergencies to know

Quick pearls

- SAH: nimodipine 60 mg 4-hourly, secure aneurysm < 24–72 h, DCI = euvolaemia + induced hypertension
- Stroke: thrombolysis < 4.5 h, thrombectomy up to 24 h (DAWN/DEFUSE-3)
- GBS/myasthenia: FVC < 20 mL/kg, NIF > −30, IVIG/PLEX
- Brain death: prerequisites, apnoea test, ancillary tests

Key point: Learn one-line management of each neuro emergency.